Physics 468/668: Statistical Physics **Spring 2019**

Lecture 5: Phase space and Liouville's theorem

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5.1 Elements of Ensemble Theory

For a given macrostate (N, V, E), a statistical system at any time, is equally likely to be in any one of an very large number of distinct microstates. When the system continually switches from one microstate to another over a span of time, we actually observe a behavior averaged over the possible microstates. Then, we expect the averaged behavior of any system in this collection as an *ensemble*.

5.2 Phase space

For a given classical system, at any time t, it may be defined by specifying the instantaneous positions and momenta of all the particles constituting the system. Let the number of particles as N, the definition of a microstate requires the information of $3N$ positions $(q_1, q_2, ..., q_{3N})$ and $3N$ momentum coordinates $(p_1, p_2, ..., p_{3N})$. Hence, we define the set of (q_i, p_i), where $i = 1, 2, ..., 3N$, as a point in a space of $6N$ dimensions. We refer to this space as the *phase space*, and the phase point (q_i, p_i) as a representative points of the given system.

$$\begin{aligned} \dot{q_i} &= \frac{\partial H(q_i, p_i)}{\partial p_i} \\ \dot{p_i} &= -\frac{\partial H(q_i, p_i)}{\partial q_i} \end{aligned} \tag{5.1}$$

where $H(q_i, p_i)$ is the *Hamiltonian* of the system. With the time passes, the set of (q_i, p_i) undergoes a continual change. Correspondingly, it forms a trajectory depending on the velocity v vector $\equiv (\dot{q_i}, \dot{p_i})$, which in turn is given by the above equation. It is not difficult to see that the trajectory of the points must remain within a limited region of the phase space, because a finite volume V directly limits the values of the coordinates q_i, and E limits the values of both q_i and p_i. In particular, the total energy of the system is known to satisfy the following,

$$H(q_i, p_i) = E. \tag{5.2}$$

This means that the trajectory needs to be restricted to the hyper-surface defined by E. If we allow the energy to vary, say $E \pm \Delta$, the corresponding trajectory will be restricted to a hyper-shell defined by the limits.

To describe the overall movements of points as a function of time, we can define a *probability density function* $\rho(p_i, q_i, t)$. This function is such that at any time t, the number of points in the volume element $(d^{3N}q d^{3N}p)$ around (q, p) is given by

$$\int \rho(p_i, q_i, t) d^{3N}q d^{3N}p \tag{5.3}$$

Similarly, any physical quantity could be described as the averaged values along the phase space.

$$< f >= \frac{\int f(q,p)\rho(q,p)d^{3N}qd^{3N}p}{\int \rho(p_i,q_i,t)d^{3N}qd^{3N}p} \tag{5.4}$$

Note that the ensemble average $< f >$ itself is also a function of time.

An ensemble is said to be stationary if ρ does not depend explicitly on time, namely,

$$\frac{\partial \rho}{\partial t} = 0 \tag{5.5}$$

Clearly, for such an ensemble, the $< f >$ will be independent of time as well. Naturally, this represents an *equilibrium* state.

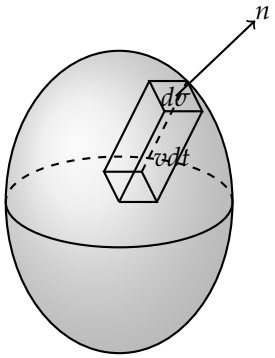


Figure 5.1: A schematic of phase space.

5.3 Liouville's theorem

Consider an arbitrary volume ω in the phase space and let the surface enclosing this volume be denoted by σ. Then the number of points in this volume increases with time,

$$\frac{\partial}{\partial t}\int_\omega \rho(t)d\omega \tag{5.6}$$

where $d\omega \equiv (d^{3N}qd^{3N}p)$.

On the other hand, the net rate at which the points flow out of ω is given by

$$\int_\sigma \rho \mathbf{v}\cdot\vec{n}d\sigma \tag{5.7}$$

Here, $\mathbf{v}$ is the velocity vector of the representative points in the region of the surface element $d\sigma$ while $\vec{n}$ is the outward unit vector normal to this element. By the divergence theorem,

$$\int_\sigma \mathrm{div}(\rho\mathbf{v})d\omega \tag{5.8}$$

$$\mathrm{div}(\rho\mathbf{v}) \equiv \sum_{i=1}^{3N}\left[\frac{\partial}{\partial q_i}(\rho\dot{q}_i) + \frac{\partial}{\partial p_i}(\rho\dot{p}_i)\right] \tag{5.9}$$

The number of points should be conserved, we have

$$\frac{\partial\rho}{\partial t}\int_\omega d\omega \equiv -\int_\omega \mathrm{div}(\rho\mathbf{v})d\omega \tag{5.10}$$

that is

$$\int_\omega\left[\frac{\partial\rho}{\partial t} + \mathrm{div}(\rho\mathbf{v})\right]d\omega = 0. \tag{5.11}$$

The necessary and sufficient condition to let the integral vanish is that the integrand itself vanish everywhere,

$$\frac{\partial\rho}{\partial t} + \mathrm{div}(\rho\mathbf{v}) = 0 \tag{5.12}$$

Let's break down the divergence term,

$$\frac{\partial\rho}{\partial t} + \sum_i^{3N}(\frac{\partial\rho}{\partial q_i}\dot{q}_i + \frac{\partial\rho}{\partial p_i}\dot{p}_i) + \rho\sum_i^{3N}(\frac{\partial\dot{q}_i}{\partial q_i} + \frac{\partial\dot{p}_i}{\partial p_i}) = 0 \tag{5.13}$$

The last group of terms vanishes identically because ,

$$\frac{\partial\dot{q}_i}{\partial q_i} = \frac{\partial^2 H(q_i,p_i)}{\partial q_i\partial p_i} \equiv \frac{\partial^2 H(q_i,p_i)}{\partial p_i\partial q_i} = -\frac{\partial\dot{p}_i}{\partial p_i} \tag{5.14}$$

Further, since $\rho \equiv \rho(q,p,t)$, the remaining term may be combined to form the total derivative of ρ as a function of t, with the results that,

$$\frac{d\rho}{dt} = \frac{\partial\rho}{\partial t} + [\rho, H] = 0 \tag{5.15}$$

where we use the *Poission braket* $[\rho, H]$,

$$[\rho, H] = \sum_{i=1}^{3N}\left(\frac{\partial\rho}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial\rho}{\partial p_i}\frac{\partial H}{\partial q_i}\right) \tag{5.16}$$

According to the theorem, the total number of systems in the ensemble is a constant. Thus the swarm of the representing points moves in the phase space in essentially the same manner as an incompressible fluids moves in the physical space!

If the density is independent of the coordinates (q,p) over the region of phase space, physically, this choice corresponds to an ensemble of systems that at all times are uniformly distributed over all possible microstates. The resulting ensemble is *microcanonical* ensemble, in which the averaged properties could be determined by the following,

$$< f >= \frac{\int f(q,p)\rho(q,p)d^{3N}qd^{3N}p}{\int\rho(p_i,q_i,t)d^{3N}qd^{3N}p} = \frac{1}{\omega}\int_\omega f(q,p)d\omega \tag{5.17}$$

A more general way of satisfying the condition is to assume that the dependence of ρ on (q, p) comes only through an explicit dependence on H,

$$\rho(q, p) = \rho[H(q, p)] \tag{5.18}$$

This equation provides a class of density functions for which the corresponding ensemble is stationary. The most natural choice of such ensemble is the one for which

$$\rho(q, p) \propto \exp[-H(q, p)/kT] \tag{5.19}$$

which is *canonical* ensemble.

5.4 Homework

1. prove that

$$\mu(N, V, T) = kT\ln\left[\frac{N}{V}\left(\frac{h^2}{2\pi mkT}\right)^{3/2}\right] \tag{5.20}$$

   and calculate μ for the following gases (He, Ne, Ar, Kr, Xe) at room temperature and ambient pressure conditions.

2. 1.12
3. 1.13
4. 1.14
5. 1.15